Strong Artificial Intelligence

D Time and Space

Guarantors of Entities

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Repetition „Reality“

- What is reality?
- Which kind of realities can be differentiated?
- Who defines the subjective reality and out of what?

Workshop „Time and Space“

- What is time? Define!
- What is space? Define!
- What are the consequences of a world without time?
- What are the consequences of a world without space?
- What medium connects time and space together?
- What entity nature has time: physical or mental?
- What entity nature has space: physical or mental?

D Time and Space

1. Time	Guarantor for Uniqueness and Singularity
2. Time Logic	Sooner or Later/Before and After
3. Space	Guarantor for Delimitability
4. Space Logic	Sphere Directions
5. Time Space	Motion
6. Time Space Logic	Is part of
7. Conclusion	Inseparable Twins

Does Change/Movement Always Need Time?	What is Time?
Yes. Think of Speed of Light c and the Special Theory of Relativity.	A Mental Concept

Is Time a Physical Entity?	What is Time Used for?
No.	To Put Movements in Order.

Without

Change/Movement?	Entities?
Change/Movement without Delay.	No Entities?

Location?	World?
No Dedicated Location for one Entity.	No World?

Time is the experience, that one entity is always percived only once.

Workshop „Time Logic“

- Where do we as humans know from, what is earlier and what is later?
- What do we need to bring the time points of events/movements into an order?
- What are natural time measurements?

Measurement Unit

Natural Unit?	What for?
One Day. The Time for One Rotation of the Earth.	Organise Temporal Order (in Work Systems).
Typical Unit?	**Needed for?**
Second. 1/3600 of a Day.	Time Logic.

2. Time Logic

Relative Logic	Absolute Logic

• Human behaviour (natural)
• Sequence of events (sooner or later)
  • If the memory of something exists while an impresson is being processed, then that which is from the memory is earlier
  • Impression or event (sequence) counter is sufficient
  • Common known synchronising anchors (events) are necessary to reconstruct a sequence chain
  • Measurement system is needed to synchrhronise multiple threads
	• Invented by humans (artificial)
	• Reference/Anchor event is needed (Gregorian calendar uses birth of Jesus Christ - AD)
	• Defined time distances provided by clocks
	• Synchronised clocks make comparisions of distributed events possible
	  • Tagging impressions with a time stamp of a synchronised clock

2. Time Logic

Absolute (clock is needed)

Two Definitions
1. Fixed Reference Point
2. Distance

Year (Sun)

Month (Year / 12)

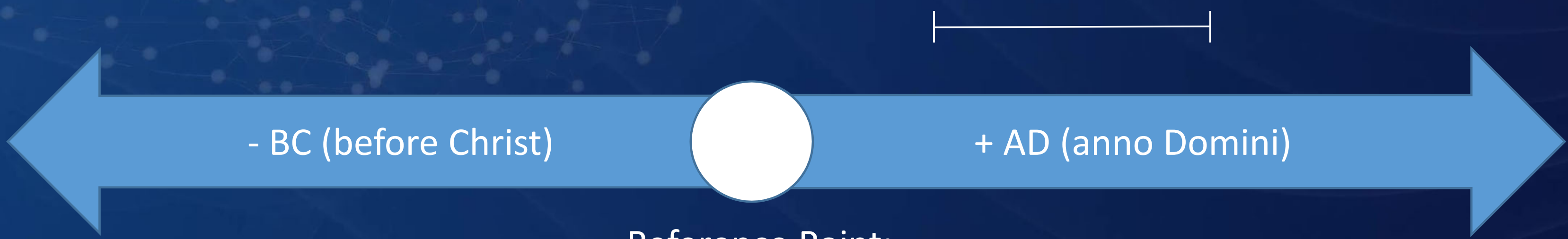
Week (Day * 7)

Day (Sun)

Hour (Day / 24)

...

Gregorian Calendar



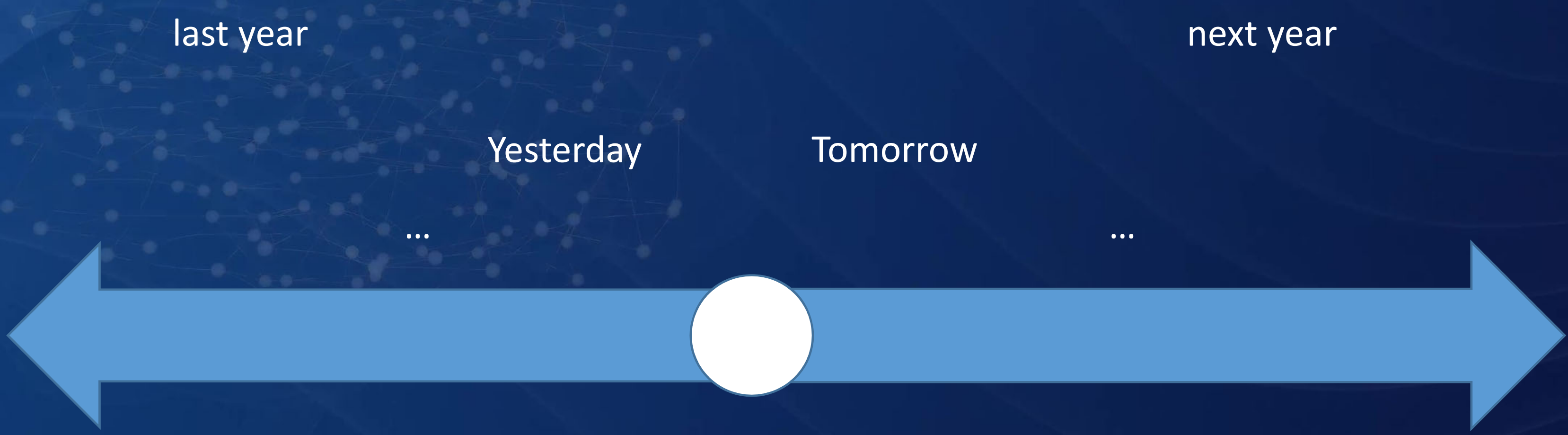
Reference Point:
Birth of Jesus Crist

2. Time Logic

Relative (no clock is needed)

Two Definitions

1. variable reference point
2. relative (often natural) distances



Reference Point:
Often „now“ or „today“

Clock: Measurement Machine

What is a Clock?	Could a Human do the Work of a Clock?
An Artificial Machine that Materialises Humans Concept of Time.	Yes.

What is the Work of a Clock?	Is Clock Work Mental Work?
Counting the Defined Measurement Units.	Yes.

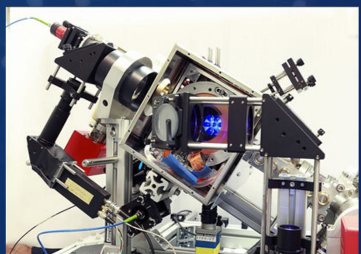
Workshop „Time Logic“

Describe the work systems of a „clock“ with three different kind of workers:

a) analog clock,

b) human,

c) atomic clock



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Workshop „Space“

- What are natural measurements units for space?
- How many dimensions has space?
- Imagine if bodies would not need space. What would happen?

Does Change/Movement Always Need Space?	What does Space Refer to?
Yes. Walking through Walls does not Work until now.	To Physical or Mental Entities.
What is Space?	**Do Mental Entities Need Space?**
A Mental Concept.	Normally Yes.

Without

Change/Movement?

Change/Movement without Restrictions.

Entities?

No Entities?

Size?

No Dedicated Size for one Entity.

World?

No World?

Space is the experience, that two entities are never at one single position and one entity is not changing its size with no reason .

Excercise

- Describe in your own words unambiguously the position of the two big screens in your discourse world „Intuition of AI“.
- Describe the position using formal variables and values.
- Extend your UML diagram by the description of the positions of the two big screens.

Measurement Unit

How Many Dimensions?	What for?
9?: 2+3 Location 1+3 Orientation	Organise Positions (in Work Systems).

Typical Unit?	Needed for?
Meter: 1/299.792.458 of the Distance Light Travelles in a Vacuum within 1 Second.	Space Logic.

4. Space Logic

Modelled on the human procedure (natural)	Invented by humans (artificial)

Modelled on the human procedure (natural)

- Relative logic
  (nested and orientated bodies)
  - All bodies are part of another body or specified space („is part of“ relationship or „part-whole“ relationship)
  - Location and Orientation relatively to a known reference point of the whole determine the Position of the part

Invented by humans (artificial)

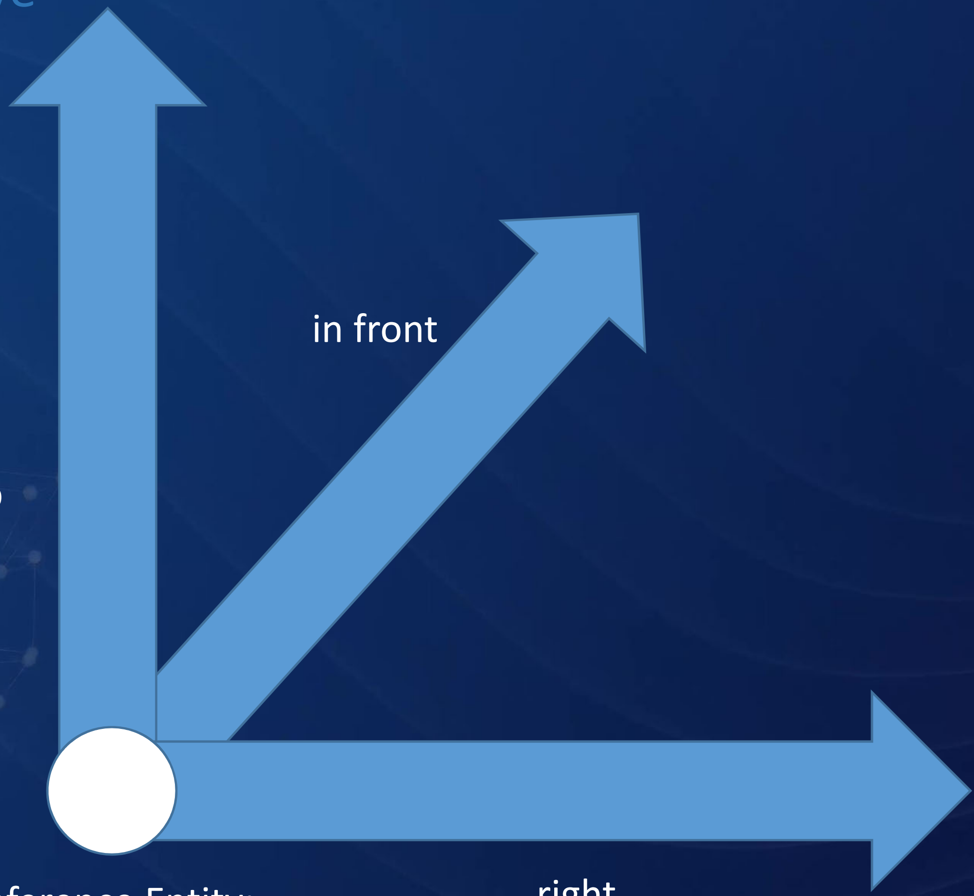
- Absolute logic
  (synchronised geopositions)
  - Abstract/artificial latitudes and longitudes on planet Earth as well as the height above the see level
  - Four points of the compass (north, east, south, west)
  - Label entities with geographical coordinates

4. Space Logic

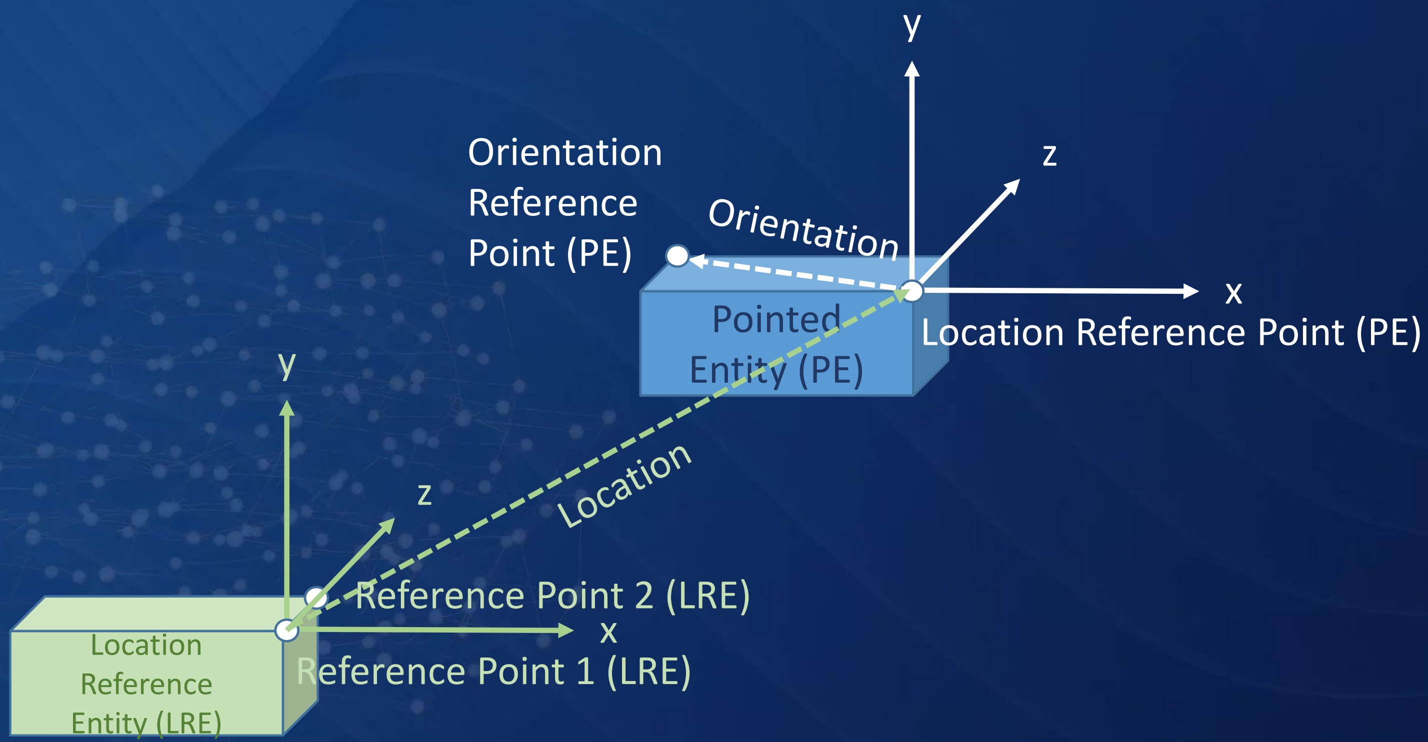
Relative

Four Definitions

1. Reference Entity
2. 2 Reference Points of the Reference Entity
3. Reference Point of the Pointed Entity
4. Location of the Pointed Entity
   in Relation to the Reference Entity
   (3 Dimensions)
5. Orientation Point of the Pointed Entity
6. Orientation of the Pointed Entity
   in Relation to the Reference Point
   (3 Dimensions)

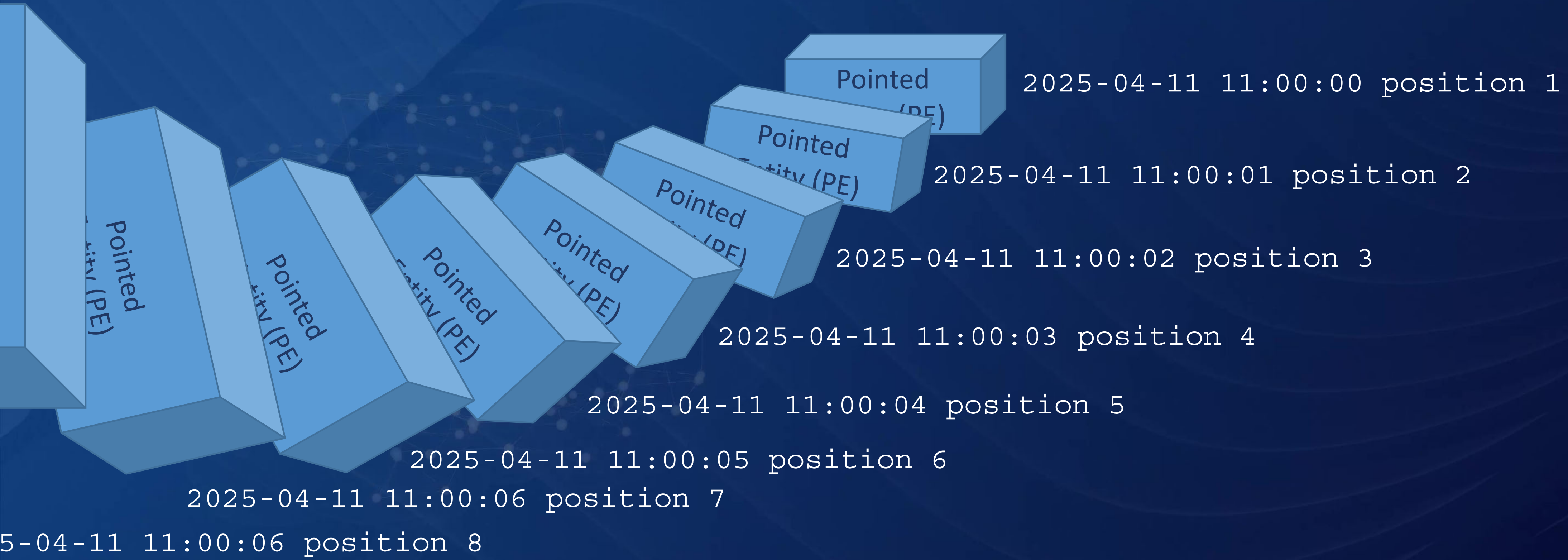


Entity Full Positioning (One Posibility)



Time (UTC)	LRE 1	LRE 2	X	Y	Z	…			

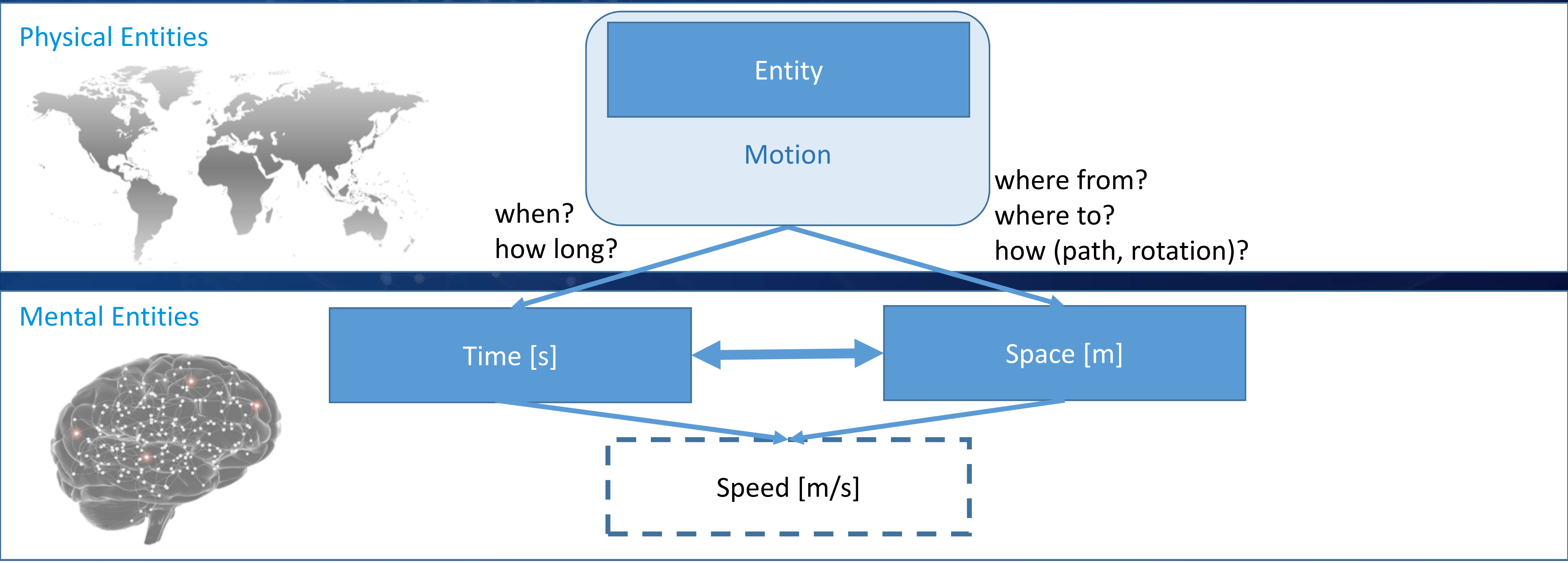
6. Time Space Logic



Time and Space are Mental Concepts, that We Use to Describe what We Perceive as Motion.

Motion Welds Time and Space together.

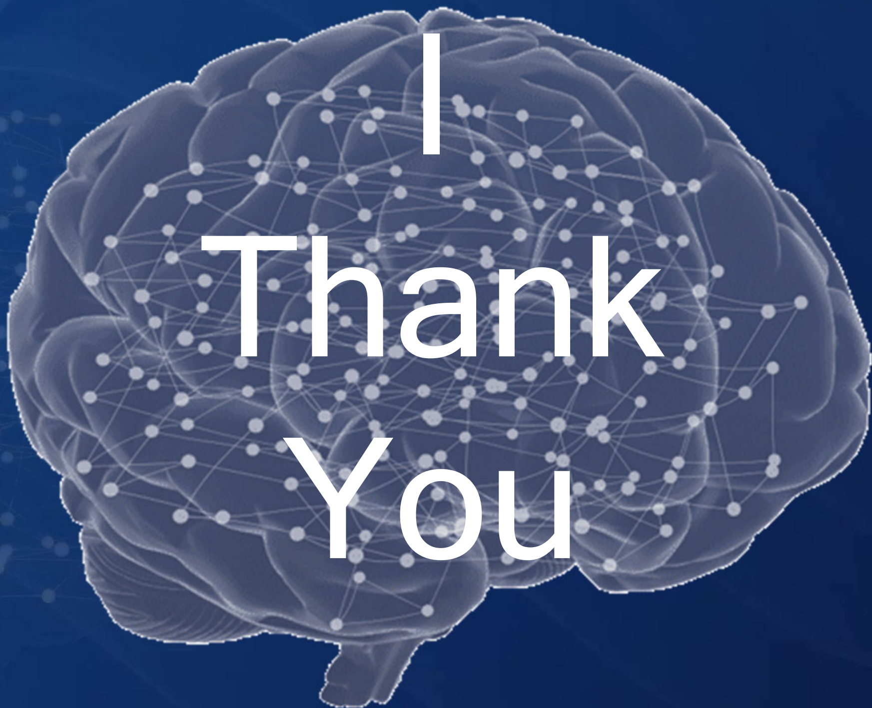
A Complete Describtion of Motion only Needs the Temporal and the Spacial Dimensions.



7. Conclusion

- Time and Space are the fundamental measures
  for motion of physical entities and inseparable.
- Mental constructions/concepts and tools are esential for developing systems that find their way in the world and are able to assist people with their work.

- Humans are (not) experts in time and space logic (memory lacks are often).
  They are experts in communicating temporal and spacial issues to each others.
- Current formal languages have no temporal and no spacial logic included, but GREBTL has (see Advanced Information Modelling).
- There is a big chance in improving these areas.
- Intelligent representing constructions/systems must have
  intrinsic temporal and spacial logic like humans have.



I Thank You